

MERCON Logistics — System Architecture

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Prepared For: MERCON Logistics Services Company

Prepared By: Starshape

1. Overview

MERCON is a comprehensive digital logistics platform consisting of three user roles — Admin, Operator, and Driver — with four dedicated applications:

Platform	Technology	Users
Driver Operations App	React Native (Android & iOS)	Drivers
Operator Mobile App	React Native (Android & iOS)	Operators
Operator Web Dashboard	React	Operators
Admin Web Dashboard	React	Administrators

All four applications communicate with a centralized cloud backend, providing seamless real-time synchronization, GPS tracking, and operational visibility.

2. Technology Stack

Layer	Technology
Mobile Apps	React Native (Android & iOS)
Web Dashboards	React
Backend	NestJS (Node.js + TypeScript)
Database	PostgreSQL (Neon PostgreSQL)
Cloud Storage	Cloudflare R2
Maps & Navigation	Google Maps
GPS Tracking	Mobile GPS (primary) + ICCES Tracker (secondary)
Notifications	Push Notifications
Web Hosting	Vercel
Backend Hosting	Railway

3. Authentication

User Role	Authentication Method
Admin	Email + Password
Operator	Email + Password
Driver	Employee ID or Mobile Number + PIN

- JWT-based authentication with role-based access control
- Sessions automatically expire after prolonged inactivity
- All user activities recorded in audit logs

4. Core Modules

The platform is organized into 16 interconnected functional modules:

#	Module	Purpose
01	Authentication & Access Management	Secure login, role-based access, session management
02	Dashboard	Operational KPIs, live activity feed, fleet status, analytics
03	Trip Management	Create, edit, schedule, monitor, complete trips
04	Route & Navigation	Pickup/delivery locations, distance & ETA, Google Maps
05	Pricing & Quotation	Route & customer rate cards, auto pricing, manual override
06	Cargo Management	Cargo type, weight, quantity, special instructions, photo verification
07	Driver Management	Driver profiles, availability, assignment, documents, license management
08	Fleet Management	Truck & trailer management, documents, insurance, maintenance schedules
09	Customer Management	Customer profiles, trip history, rate cards, billing records
10	Live Tracking	Mobile GPS + ICCES tracking, trip progress, ETA, route monitoring
11	Driver Workflow	Guided trip execution: start → cargo → navigate → arrive → deliver → complete
12	Notification & Communication	Push notifications, emergency alerts, WhatsApp trip sharing
13	Reports & Analytics	Daily/weekly/monthly reports, revenue, fleet, KPI analytics
14	Administration	User & role management, system configuration, audit logs
15	Document Management	Driver & vehicle documents, permits, delivery proof, expiry tracking
16	Platform Services	Cloud database, APIs, security, backup/recovery, performance monitoring

5. Trip Workflow

The transportation lifecycle follows this standardized flow:

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Customer Request
↓
Operator Creates Trip (Status: Created)
↓
Route Selection (Pickup & delivery via Google Maps)
↓
Cargo Information Captured
↓
Trip Scheduled (one-time, daily, weekly, or monthly)
↓
Automatic Pricing (system suggests based on route & customer rate cards)
↓
Assign Vehicle (Status: Assigned)
↓
Assign Driver → Trip pushed to Driver App + shared via WhatsApp
↓
Driver Opens Trip (Status: Ready to Start)
↓
Driver Taps "Start Trip" → Mandatory Cargo Photo Upload (Status: Cargo Verification)
↓
Trip Officially Begins (Status: In Transit) → Navigation opens automatically
↓
Live GPS Tracking (continuous sync with Operator Dashboard)
↓
[Optional] Emergency / Trip Halt (Status: Emergency/Halted → Operator notified immediately)
↓
Arrival at Destination (detected via GPS geofencing, 500m radius) (Status: Arrived)
↓
Mandatory Delivery Photo Upload (Status: Delivery Verification)
↓
Trip Completed (Status: Completed) → Reports & customer history updated

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Trip Status Lifecycle

Status	Description
Created	Trip record initiated, not yet assigned
Assigned	Driver and vehicle assigned to the trip
Ready to Start	Driver has acknowledged/opened the assigned trip
Cargo Verification	Driver uploading mandatory cargo photos
In Transit	Trip actively in progress, GPS tracking live
Emergency / Halted	Trip paused due to emergency or halt
Arrived	Driver arrived at destination (GPS geofence triggered)
Delivery Verification	Driver uploading mandatory delivery photos
Completed	Trip successfully finished, records updated
Cancelled	Trip cancelled by operator

6. GPS Tracking

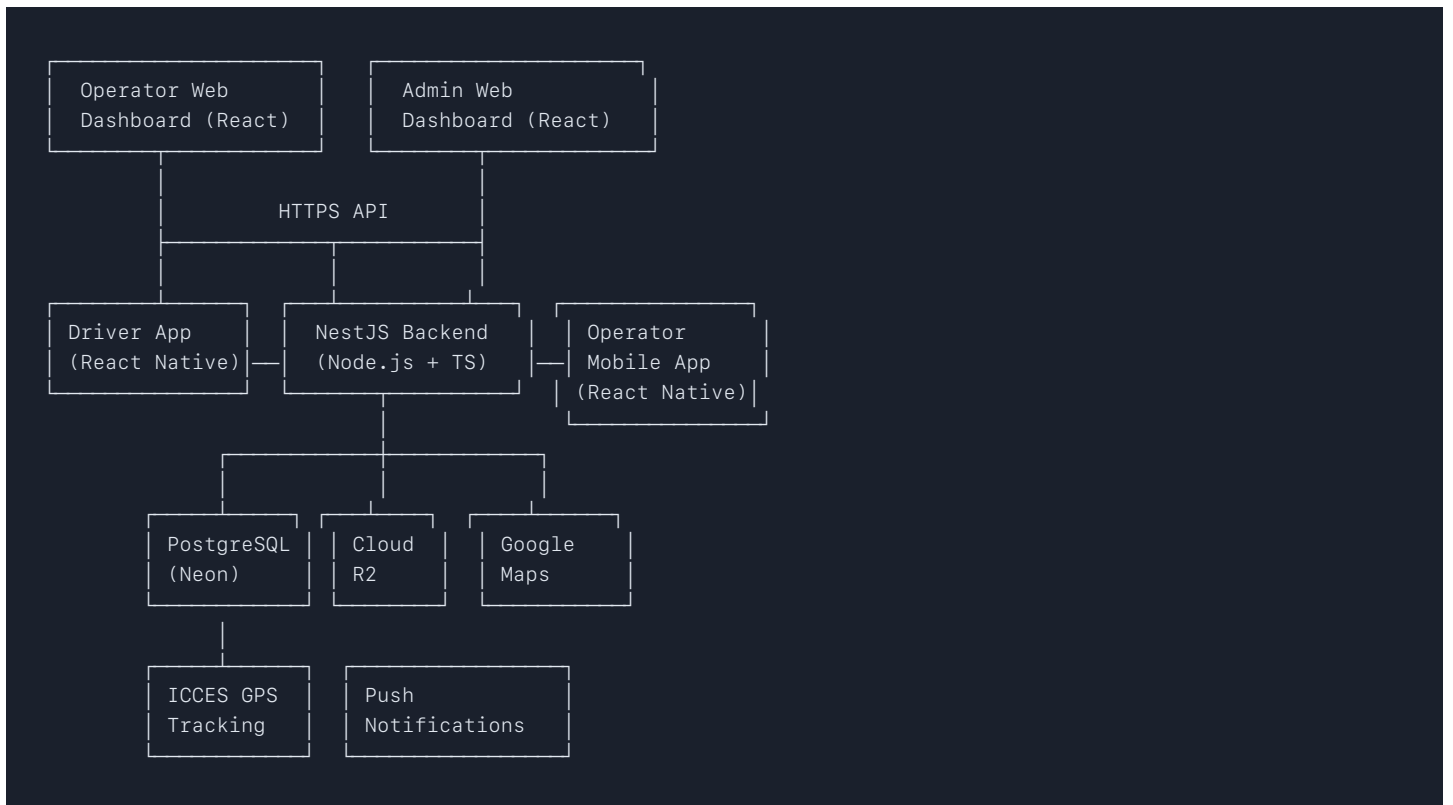
Source	Role	Description
Driver Mobile GPS	Primary	Real-time location from the Driver App via Google Maps
ICCES Vehicle GPS	Secondary	Hardware GPS device on the vehicle via ICCES API

- GPS tracking begins automatically when a trip starts and ends when it completes
- Location updates synchronize continuously with the Operator Dashboard
- Arrival is detected automatically via GPS geofencing (default radius: 500 meters)
- Fallback: Manual "I Have Arrived" button if geofence doesn't trigger

7. Security

Measure	Description
HTTPS	All communication encrypted in transit
JWT Authentication	Token-based authentication with expiry
Password / PIN Hashing	Secure credential storage
Role-Based Authorization	Permissions enforced per role across all apps
Audit Logs	All critical system actions logged
Upload Validation	Media file validation and sanitization
API Rate Limiting	Protection against abuse

8. Architecture Diagram



9. Scope Boundaries

In Scope — MVP

- Authentication & user management (Email/Password for Admin/Operator, Employee ID or Mobile Number/PIN for Driver)
- Trip management with full lifecycle (10 statuses)
- Pricing & quotation management
- Live GPS tracking (Mobile GPS + ICCES)
- Driver guided workflow with photo verification
- Notifications (push + WhatsApp sharing)
- Reports & analytics
- Fleet, driver, customer management
- Document management with expiry tracking

Out of Scope — MVP (Future Enhancements)

- Offline mode with auto-sync
- Automated billing & invoicing
- Customer self-service portal
- AI-based route optimization
- Dynamic pricing engine
- Digital signature for delivery

- Driver performance scoring
- Fuel & fleet health analytics
- Predictive vehicle maintenance
- IoT sensor monitoring
- Multi-language support
- ERP / CRM & 3rd-party integrations
- Multi-company / multi-branch operations